

Haily J

IoT Engineer

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PROFESSIONAL SUMMARY

Expert in connected devices, embedded systems, MQTT, cloud IoT platforms, and sensor integration, delivering secure, low-latency, scalable, fault-tolerant, and high-performance solutions. Skilled in real-time data analytics, edge computing, device-to-cloud communication, Celery, Redis and optimizing device performance for diverse industrial and consumer applications.

WORK EXPERIENCE | 6 Months

Associate IoT Engineer | [Foxtech Private Solutions](#)

11/2025 - Present

AI Rover

- Worked as an assistant on building an autonomous robotic system using Raspberry Pi 5, running two AI models concurrently for object detection and classification, integrating sensor controls and optimized with multi-threading for real-time performance.
- Implemented multiple autonomous modes with real-time live camera feedback, IMU sensor integration, and compass data for precise orientation, supporting full offline operation.
- Managed device control and sensor data using threading and FastAPI endpoints with async/await, achieving sub-100ms response latency.
- Optimized navigation algorithms to improve obstacle avoidance by 30%, ensuring seamless hardware-software integration with motors, sensors, and IoT modules.

PERSONAL PROJECTS

IoT Smart Scheduler <https://github.com/shelj-dev/IoT-smatscheduler>

- Developed an IoT-based smart scheduling system using microcontrollers and cloud integration. Implemented real-time data communication using MQTT and REST APIs. Built a responsive web dashboard for task scheduling, monitoring, and device control.
- Integrated sensor/device data with backend services and database management. Enabled Wi-Fi connectivity and remote access for smart automation features. Used HTML, CSS, JavaScript, and Django backend technologies for full-stack IoT development. Performed device testing, API testing, and troubleshooting for reliable system performance.

Technologies Used: IoT, Embedded Systems, Django, HTTP APIs, HTML, CSS, JavaScript, MySQL, Raspberry Pi pico 2w, GitHub

SKILLS

Programming Languages: Python, JavaScript, HTML, CSS, SQL

Wired & Wireless Technologies: I2C, ADC, SPI, UART, CAN, Ethernet, Wi-Fi, Bluetooth

Networking Technologies: TCP/IP, DNS, DHCP, HTTP, MQTT, SSL/TLS

Development Boards: Micro Python, Arduino, Raspberry Pi, ESP8266, Sensors (IMU, LiDAR)

Backend: Django, Django REST Framework (DRF), MQTT

Frontend: React.js, Tailwind CSS

API & Documentation: RESTful APIs, Swagger UI, Postman

OS & Cloud: Amazon Web Services, Linux, Windows, IoT Cloud Platforms (ThingSpeak, MQTT brokers, Blynk)

Tools & Version Control: Git, GitHub, CI/CD pipelines, Version Control

EDUCATION

Bachelor of Technology in Computer Science

05/2023 - 06/2026

Nesamony Memorial Christian College

Marthandam

INTERESTS

- Surf Tech on YouTube, IOT